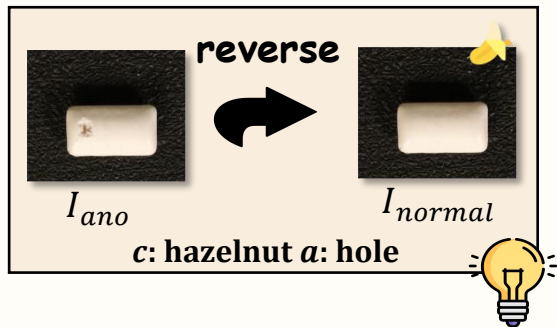
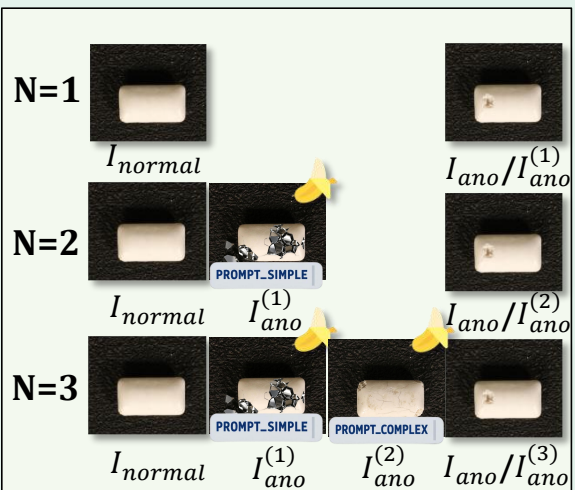


Reverse Synthesis

Step 1 Normal Image Reconstruction



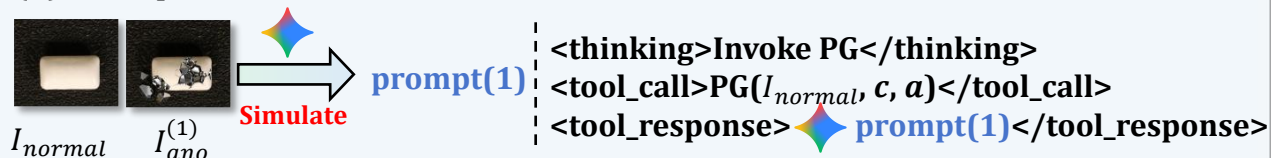
Step 2 Multi-step Anomaly Preparation



Trajectory Construction

Step 3 Trajectory Construction

(1) Prompt Generation



for $t = 1$ to N do

(2) Image Generation

```
<thinking>🌟 (refine prompt for step  $t$ ) & Invoke IG</thinking>
<tool_call>IG( $I_{normal}$ , 🌟 simulated prompt( $t$ ))</tool_call>
```

```
<tool_response> $I_{ano}^{(t)}$ :  or  or  </tool_response>
```

$I_{ano}^{(1)}$ $I_{ano}^{(2)}$ $I_{ano}^{(3)}$

(3) Quality Evaluation

$I_{ano}^{(t)}$ $\xrightarrow{\text{QE}}$ **score(t), review(t)**

```
<thinking>Evaluate Anomaly Quality of  $I_{ano}^{(t)}$ </thinking>
<tool_call>QE( $I_{normal}$ ,  $I_{ano}^{(t)}$ )</tool_call>
<tool_response>🌟 score(t), review(t)</tool_response>
```

(4) Knowledge Retrieval (Optional)

c, a $\xrightarrow{\text{KR}}$ **Knowledge**

```
<thinking>🌟 Retrieve Knowledge of (c,a)</thinking>
<tool_call>KR(c, a)</tool_call>
<tool_response>🌟 Knowledge</tool_response>
```

Step 4 Mask Generation



Step 5 Termination Signal

```
<thinking>🌟 Task Completed</thinking>
<answer>🌟 Summary</answer>
```